

Futures Narrative:

How Agentic AI can strengthen Predictive AI in the apparel
retail industry and improve planetary health

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M7: Design for Responsible AI

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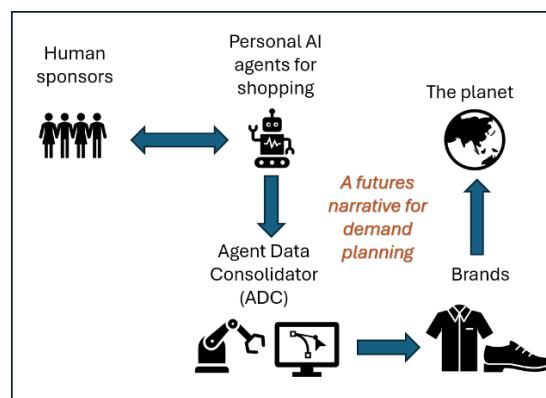
I. Introduction

This futures narrative will demonstrate that a widespread transition to using personal AI agents for shopping in the next decade (2026-2036) holds the potential to refine how demand forecasting is done for goods in the apparel industry by unveiling new levels of granularity with more reliability. Since the apparel industry is one of the top planetary pollutants with its taxing production processes and global supply chains, any improvement in demand planning will translate into lesser overproduction and environmental contamination, which will represent a relief for our planetary health.

The path to achieve this benefit is rooted in Responsible AI innovation and the application of Anticipatory AI ethics while steering the process. In first position, Responsible AI innovation is needed because, as this paper will show, the emergence of a new futures persona (the algorithmic consumer) will predispose the creation of a new futures infrastructure tool (the Agentic Data Consolidator) to make use of the enormous data from each personal AI agent on consumer shopping requests. The personal AI agents and the new AI tool have to be both designed under Responsible AI principles to be effective for their purpose, while the entire sociotechnical environment will need Anticipatory AI ethics to govern the transformation process.

This paper first begins by providing an overview of demand planning in apparel retail companies and demystifies what Predictive AI means in this business function and how it has evolved. It then pinpoints the specific methodological opportunity where agentic commerce can provide more granular data and improve demand forecasting. The paper then switches to developing a futures narrative by introducing an anticipated new futures persona (the algorithmic consumer), an opportunity for a new futures infrastructure tool (the Agentic Data Consolidator), and a futures timeline for these developments with governance structure ideas. In the final part, it translates how demand planning improvements will reflect in industry operations and improve planetary health.

Illustration of all main agents in this futures narrative:



To support this futures narrative, the author is applying his professional experience from Levi Strauss & Company (Levi's) to illustrate certain methodologies and data practices. The mature profile of Levi's as a global maker of denim garments with 170+ years of history provides useful context for situating this proposal on business processes that have undergone extensive development and proven durable.

The futures framework developed in this paper aims to answer the Anticipatory AI questions of "What will happen?" (the emergence of the algorithmic consumer), "How can we use that?" (the creation of the Agentic Data Consolidator to harness the data from the personal AI agents) and "Why would we do it?" (to improve planetary health via reduction in apparel overproduction and associated environmental damages).

II. Key terms in the context of this paper

Apparel retail industry: collection of companies selling apparel products (clothes) with global supply chains where materials sourcing and production is geographically far removed from the actual consumer.

Demand planning and forecasting: Demand planning is the overall business function, while demand forecasting is an underlying activity responsible for producing estimates for future sales of apparel goods based on statistical analyses from past sales, pricing data, and consumer trends and surveys.

Predictive AI in demand forecasting: combining 1st party data (company's own data) with 3rd party data (external data) to create a more holistic market view and expose the computational models to diverse data sets (move away from the self-prophecy of 1st party data). Machine learning methods are applied for big data analyses.

Agentic AI: systems that act with a degree of autonomy towards a goal. Able to plan, make decisions, take actions, use tools, adapt to feedback.

Personal AI agent: a *user-centered implementation* of agentic AI, designed to run on PCs and other digital devices.

Algorithmic consumer: A new term for the apparel consumer who interacts with brands and marketplaces through personal AI agents, which programmatically do survey-of-the-land and collect matching options for their "human sponsor".

Agent Data Consolidator: An infrastructure tool in the AI ecosystem that stores and aggregates data on consumer shopping requests from millions of personal AI agents. Provides brands with data for the *real total market demand* for apparel goods.

Planetary health: The reduction in supply chain negative externalities from improvement in apparel demand forecasting that minimizes excess production and supply. Those negative externalities include overproduction (which uses hazardous chemicals and environmentally contaminating processes), emissions from unnecessary raw and finished goods shipments, inventory buildup, transfer to landfill and dumping sites, overall environmental pollution.

III. Overview of current demand planning practices in apparel retail

In the apparel retail industry, demand planning is a crucial analytical exercise that has consequential effects across the entire supply chain of a retail company¹. Once an apparel retail company (such as Levi's or H&M, for example) makes estimations for the demand of its products (typically done by season), those numbers guide the entire manufacturing, shipment, inventory, and allocation processes, i.e. the entire path from the production factory to the shelves in the stores. Thus, having the ability to correctly forecast product demand becomes a core requirement for maintaining a lean and efficient supply chain. Improving this forecasting ability will not only improve the financial balance sheet of the company in question but affect the entire planetary health in a beneficial way because the majority of those supply chains are global, massive in size, with highly taxing to the environment operations².

Even though every apparel retail company strives to produce forecasts as accurately as possible, this remains a challenging task because human behavior with respect to fashion is very fickle³ with personal styles often changing because of surrounding social phenomena. In that way, demand planning is akin to a prediction market in which the payoffs affect our planetary health - if the forecasts overshoot actual demand, this will lead to overproduction (which uses hazardous chemicals and environmentally contaminating processes), emissions from unnecessary transfers and shipments, inventory buildup, potential landfill dumping, and overall environmental pollution. If, on the other hand, the forecasts underestimate demand, this may trigger emergency production requests to supply the missing quantities, which can throw off normal business operations and introduce noise and chaos in the system.

In practice, demand planning accuracy may differ a lot by product. One study⁴ reported that for high-volume, stable products (e.g. straight-fit jeans, basic tees), accuracy

tends to be high: 85% to 95% as large volumes smooth out random variations and demand patterns are predictable. For fresh products with weather-sensitive (seasonal) demand (e.g. summer dresses), accuracy is a little lower between 70% to 80% as there is inherent volatility in the selection choices by the consumers, while for slow-moving products with intermittent demand (e.g. belts, vests, overalls, etc), accuracy can drop to between 50% and 70%. If I may cite a KPI metric from Levi's where ~80% of the business is from high-volume, stable jeans products⁵, the goal for demand planning accuracy is 92%, which would represent the high-end of demand accuracy for any apparel retail company.

Illustration of product types and demand forecast accuracy:

<i>Product type</i>	<i>Demand forecast accuracy</i>
High-volume products with stable products	85-95%
Fresh products with weather-sensitive demand	70-80%
Slow-moving products with intermittent demand	50-70%

Because of this complexity, demand planning is a decision science function that is often described as an exercise of art and science between humans and computers⁶. Computers enable us to run quantitative models that use multiple regression modeling with time-series from past sales to evaluate the weights for each product's performance and extrapolate future sales⁷. To do that, an apparel retail company would use its own (1st party) sales data, which is typically found to be the greatest predictor. However, this methodology is a form of self-prophecy where the company is only analyzing the universe of its own customers and planning products for the same (sometimes monolithic) consumer group. This works quite well to bring cohesiveness and focus to the business and marketing strategies, but from responsible innovation standpoint it is not an equitable way of "hearing all voices" as certain consumer groups get excluded from future demand by way of the mathematical weights coming out from the regression models. In this way, demand forecasting from 1st party data carries inherent bias.

For example, the largest excluded group in fashion is oftentimes the "Plus" size consumer⁸ (despite actual demand⁹) because many brands only start with a limited set of offerings for this category, the sales for which never grow in the internal data set to signal for more demand (in this otherwise data-driven decision). So the exclusion is getting self-perpetuated as the Plus Size consumer may find the selections too few, basic, or unsuitable, and abstain from buying. Other excluded groups could include specialty

workwear groups (e.g. workers who need thermal clothing), LGBT groups (for whom “Pride Month” collections have turned into political fights¹⁰, or in another example, where drag queens need stiletto shoe size 14), disabled people, and more. Thinking from this frame, creating apparel merchandise for everyone can be a difficult and costly process, but if there is a more reliable signal that *total market demand* exists, it can become financially attractive in net-present-value for companies to adjust their production processes.

Illustration of the self-prophecy trap and possible discriminated groups:

Self-prophecy in demand planning: the universe of a brand's own consumers

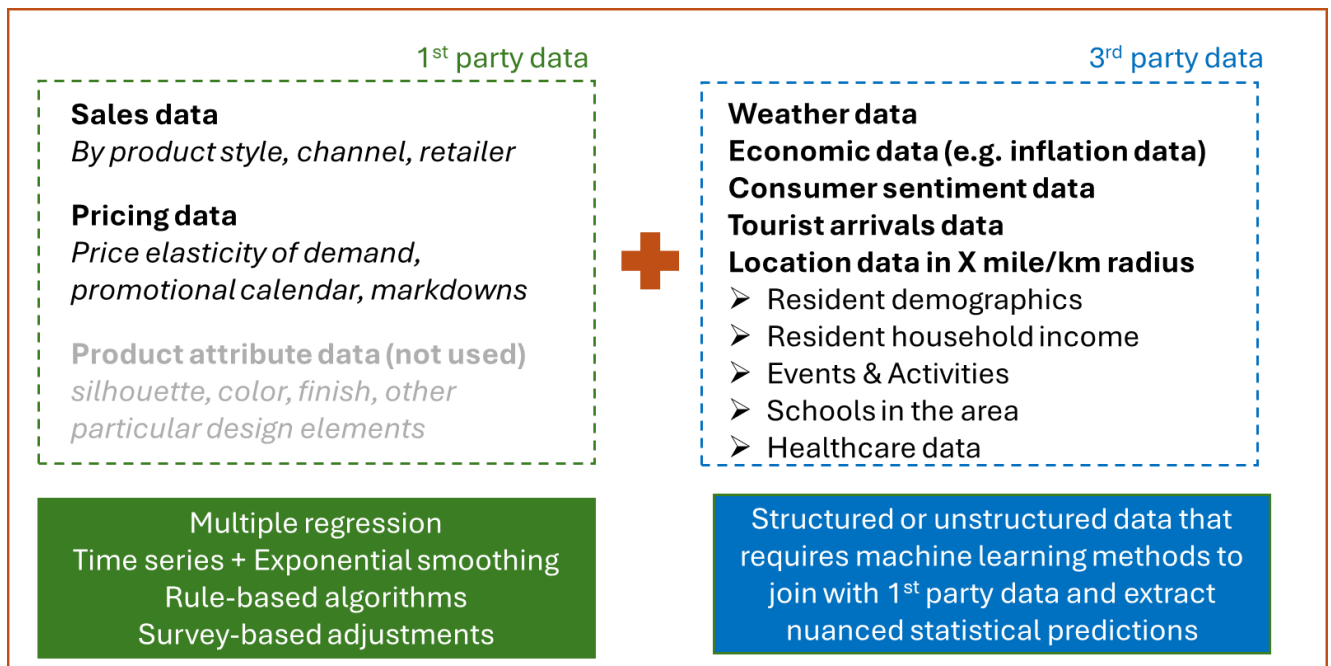


Excluded group examples:



This is why, with the vast proliferation of data from all sorts of domains, apparel retail companies are trying to escape the self-prophecy trap in their demand planning regression models by enriching those modes with external data sets. Those external data sets can range from resident demographics of the consumers around their stores to tourist arrivals data to weather forecast data to inflation and economic data to home value prices in the area to consumer sentiment data to health spending data in the area, any data set that has localized market dimensions to bring more diversification and contextual differentiation in the computational models. This data enrichment practice has converted demand planning into a Predictive AI process (Figure 1) where machine learning techniques are now employed to handle the processing and analysis of all this disparate data.

Figure 1: Demystifying Predictive AI in demand planning



= Predictive AI

The goal has become to forecast demand in the context of the entire market, not only the company's product portfolio. When demand forecasts are produced, they also have to align to the product hierarchy. Forecasts can be done at very granular levels by channel, product finish type, and even size, but demand planning teams in apparel companies often prefer to take a "middle position" and submit estimates by channel and product style (one level up from product finish) (Figure 2). The reason for this is that at the most granular level, there is randomness (in statistical sense¹¹) that can be misleading and create outsized estimations.

Figure 2: Demand forecasting illustrated in view of product level hierarchy

Currently

Product level hierarchy	Example 1	Example 2	
Product category	Top	Bottom	 This is the “middle” level at which demand forecasting is done with greatest total accuracy.
Product type	Fleece hoodie	Jeans	
Product style	Relaxed fit	Slim fit	
			Demand planning team
Product finish	Pink	Distressed Midnight Blue	Merchandising teams allocate the totals to finishes according to their observations of market trends. Sizing uses percent allocations.
Product attributes	Graphic print	Zipper fly, wide ankle	
Product size	M	32x32	
			Merchandising team

The above is done by channel (wholesale, specialty (e.g. army), own-and-operated, e-commerce)

For example, if I want to do a flash mob stunt on the street and buy 20 pink hoodies for my dancers, the more reliable demand information is not that I needed hoodies in pink, an event that may not repeat next year, but that I needed hoodies during that season. If the demand planning team did the forecasting at the most granular level and produced a demand signal that next year the company needs to stock 25 more pink hoodies, this granularity will create excess supply if another creative chooses to upstage me with a flash mob requiring 30 hoodies in any other color but pink for differentiation. This excess supply will turn into unsold goods, which the company may try to flush through markdown sales or send off to the off-price market. At the end, those hoodies may very well end up in textile dumps taking decades to biodegrade. Thus, wrong mathematical methodology may add to environmental pollution for the planet. This is why, because of this reality of non-repeatable, randomly distributed events among fashion shoppers, demand planning for large apparel retail companies is more secure to be done at a middle level, that of a product style (e.g. fleece hoodie) than at a product finish one (e.g. fleece hoodie in pink).

The central narrative in this paper will depict a futures scenario where personal AI agents would be able to supply this most-granular information with predictability, which will reflect *total market demand* and will in turn make demand planning estimates sharper, leading to more efficient supply chains and environmental relief for the planet (Figure 3).

Figure 3: Future scenario for demand forecasting with input from Agentic AI

Future Scenario with Agentic AI

Product level hierarchy	Example 1	Example 2	
Product category	Top	Bottom	Demand planning team operates as usual at this level but gets additional input from Agentic AI for lower-level demand.
Product type	Fleece hoodie	Jeans	
Product style	Relaxed fit	Slim fit	
Product finish	Pink	Distressed Midnight Blue	This is the information that a personal AI agent can supply about individual preferences with greater reliability.
Product attributes	Graphic print	Zipper fly, wide ankle	
Product size	M	32x32	

- Bottom-up forecasting vs middle-up forecasting
- Increased accuracy, leaner supply chain

One of the merits in this art and science demand planning process is that humans still exercise considerable oversight over the estimations produced by the computers as human managers review, amend, approve, and send those estimations to the next chain of command that starts the production process. Thus, it could be argued that the current demand planning process is well-designed at the top for human oversight, but could be improved at the bottom, which is where Agentic AI can supply granular data from millions of consumers about their near-term product needs.

IV. Futures persona: the algorithmic consumer via personal AI agents

In its “State of of Fashion” 2026 annual report¹², McKinsey & Company highlights that “in the years ahead, autonomous AI shopping agents may act on their [consumers’] behalf, completing tasks from monitoring prices to buying products”. This will signify a new era in which we might see the emergence of a new persona: the algorithmic consumer. The Harvard Journal of Law and Technology already recognized it as early as 2017, publishing an extensive paper¹³ on the subject. The incentives behind the adoption of personal AI agents for shopping center on the premise that such AI agents can significantly reduce search and transaction costs, help consumers overcome biases to enable more rational and sophisticated choices, and create or strengthen buyer power.

However, the personal AI agents would not do this in a vacuum but would already have a wealth of information about their “human sponsor”. Those personal AI agents would have access to all digital touchpoints, history, and personal data of their human sponsor: data points such as previous purchasing history, exact sizing information, future calendar on which events may be marked down, “likes” on social media, style preferences, and their

human sponsor's current inventory of clothes. Knowing the entire microcosm of the consumer would enable the personal AI agents to engage in very targeted shopping moves, sometimes knowing far in advance what their human sponsor will need at a future date.

To illustrate this, I will provide two futures examples:

Example 1: Rafael, 26 years old in Barcelona Rafael is a sports person. He will run in the Barcelona marathon in March and has purchased tickets to watch the ATP 500 tennis tournament in Barcelona in April. Rafael will need new shoes after the Barcelona marathon to start breaking them in for his next marathon (his old ASICS ones already have 500km on them, which is the limit they were designed for). He will also need new sunglasses for the summer. The week when he will be all day out at the ATP 500 tournament in Barcelona is predicted to be chiller than usual so he will also need a windbreaker jacket and a nice Ralph Lauren polo shirt for a timeless look. Raphael's personal AI agent knows his size and style preferences as well as his budget. It has sent a signal to ASICS and Ralph Lauren three months ahead of time. As a result, Rafael gets 10% "early order" discount. As for the sunglasses and the windbreaker jacket, Raphael is not sure what he wants so his personal AI agent accesses a marketplace for algorithmic shopping to find items that match Raphael's aesthetic and budget. Raphael will approve them and the AI agent will order them.	Example 2: Maria, 40 years old in New York Maria gets a wedding invitation from one of her friends from college, who is getting married in Miami, FL. The invitation is sent 6 months ahead of time. Maria decides that she would need a cocktail dress and an evening gown. She wants to update her wardrobe for the occasion (plus, you never know who you will meet at weddings!). Because a trip to Miami, FL, will be already expensive, Maria is considering getting a cocktail dress from Zara and an evening gown from Nordstrom. She has previously shopped for a pair like that. Her personal AI agent sees the purchase history and has a conversation with her about what styles she may prefer, suggesting recent likes that Maria has made on social media. Buying a dress and a gown is more complex than other everyday items, so her personal AI agent reserves several dresses at Zara and several gowns at Nordstrom for her at prices that are reasonable for her wallet. Those dresses will be automatically carried to a dressing room for Maria to try at once at the time when she comes for her shopping appointment. Her AI agent books the appointment.
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These examples illustrate how easily personal AI agents may integrate with human lives and take the initiative for apparel retail shopping. I also portrayed some tactical

elements in the design of the future shopping experience to enable them to be incrementally assistive and act as connectors:

- (1) Early purchase order is rewarded with some discount (e.g. 10%)
- (2) Bookings of fitting rooms with pre-selected items can be arranged for max efficiency
- (3) Online marketplaces are set up for AI agents to browse and bid for

However, such integration also opens the question how the human sponsors' rights will be affected if AI agents will be making selections and even acting on their behalf. It is of central importance to then conduct an AI Anticipatory review for this human-computer interaction scenario (Table 1) and answer the question of “**what Responsible AI properties the personal AI agent should be equipped with in this futures scenario**”. The human sponsor will retain final approval (oversight) on the shopping process, but the AI agents would represent him or her on various marketplaces and brand platforms.

Table 1: Responsible AI requirements for personal AI agents for shopping

Property	Level of importance	Comments
#1 Transparency	Very important	It is very important for the personal AI agent to be transparent in communicating what it has access to (e.g. shopping sites, review forums, AI agent marketplaces) and how it operates (e.g. selects and waits for human review & approval or gets authorized to take decisions on its own in certain cases)
#2 Explainability	Very important	It is very important for the personal AI agent to understand the dimensions of its human sponsor's need and be able to explain why it made certain selections to match those needs.
#3 Security	Very important	It is very important for the personal AI agent to maintain security of its human sponsor's data because any jailbreak exposes the human sponsor to risks when personal data is stolen.
#4 Privacy	Very important	It is very important for the personal AI agent to maintain privacy for its human sponsor and not reveal his or her identity because malignant parties may take advantage of any identity information for nefarious purposes.
#5 Reliability	Important	It is important, but not a must, for the personal AI agent to perform without failure within the given timeframe (for example, failure on “Find me new

		shoes tomorrow” may not a critical design requirement).
#6 Resilience	Important	It is important for the personal AI agent to stay resilient if any internal or external circumstances change such when a marketplace changes payment terms, return policies, or privacy and security policies.
#7 Fairness	Neutral	Since it serves a single human sponsor, the personal AI agent selects choices that match this person’s needs in a fair way. The human sponsor can retain the final decision and implicitly validate the fairness of the work by the AI agent.
#8 Interpretability	Neutral	It is not important for the personal AI agent to understand the relationship between the needs of his or her human sponsor. If the human sponsor is looking for both a ballet tutu and a leather biker jacket, the personal AI agent can remain neutral to the origin of this demand.

Thus, the technological horizon for the personal AI agents should be designed with the above considerations for Responsible AI to make them truly effective and prepare them for a secondary role in becoming transmitters of data for bigger purposes, such as demand planning.

V. Futures tool: the Agentic Data Consolidator

Since each personal AI agent will have information about the shopping intentions and future orders of its human sponsor, the next logical question is how this data could be collected so that its aggregate form can become useful for business applications such as demand planning. Furthermore, since personal AI agents will run on a diverse set of devices (PCs, phones, tablets, watches, etc), there is a business opportunity to capture this fragmented but collectively exhaustive data into one central system of record. This is where I propose a tool called “**Agent Data Consolidator**” (ADC) to act as enabler for that.

The ADC will be a cloud-based storage repository, a data lake, which will collect and update data from millions of personal AI agents about their human sponsors’ near-term shopping needs at very granular level – by product finish and size. The ADC will have an intake form, which the personal AI agents will fill out. Every time their human sponsor mentions a shopping need, the personal AI agent will connect to the ADC and log a shopping request (a unit of demand). On the other side, brands can monitor all demand

entries and send proposals directly to the personal AI agent, which can take action to present them to its human sponsor or disregard them because of certain criteria.

The ADC will fundamentally run on the principle of anonymity because the essential data needed is not *who* made the order, but *what* the order is and *where* it is located at postal code (zip) level. Thus, the ADC will be disabled from taking directly the name, age, street address (it will only take the postal (zip) code) or any personal data about the consumers, as well as any payment information. It will, however, host the ID of the personal AI agent that will be in charge of managing the entry. In this way, the human sponsor will not be directly traceable from the ADC (but could be traceable through the ID of his or her personal AI agent – this is why the Responsible AI design for the personal AI agent is of absolute importance for privacy and security). This setup is by intentional design to prevent the ADC from becoming a surveillance tool for individual consumers and their preferences when the only thing it should be a “surveillance tool” of is *total market demand*.

So the role of the ADC is not to facilitate order fulfillment (which will still be carried out through normal channels and occur on the brands’ point-of-sale systems), but to aggregate data for demand planning and give brands the opportunity to surface items to the personal AI agents that match their human sponsors’ shopping requests. Transactions will not be carried through the ADC.

An example of a shopping request that would be stored in the ADC for Rafael’s running shoes and polo shirt is presented in Table 2 below. Different apparel brands may use different product hierarchies and nomenclature to designate and describe their products, but the intake form in the ADC will be mapped to each brand’s peculiarities through an alignment and harmonization dictionary (created by AI for each brand).

Table 2: Illustration of data tables stored in ADC from personal AI agents for shopping

	(for Rafael)	(for another person)	(for another person)	(for another person)	(for another person)
AI Agent ID	A5638260473	A5635815473	A5630063473	A5638273523	A5638864073
Postal Code	08010	40276	15270	38679	22633
Category	Accessories	Accessories	Accessories	Accessories	Accessories
Type	Shoes	Shoes	Shoes	Shoes	Shoes
Style	Running	Running	Running	Running	Running
Width	Wide	Standard	Standard	Wide	Standard
Color	White	Neon blue	Orange	Pink	White
Cushion	Regular	Extra	Regular	Extra	Regular
Pronation	Neutral	Overpronate	Neutral	Neutral	Overpronate
Size	13	12	14	13	12
Budget	\$80-\$100	\$120-\$140	\$120-\$140	\$80-\$100	\$100-\$120
Date Needed By	06/31	07/15	09/15	07/31	05/15

**Product
finish info
& size**

	(for Rafael)	(for another person)	(for another person)	(for another person)	(for another person)
AI Agent ID	A5936560473	A5600545473	A5630063383	A5664773523	A5644784073
Postal Code	08010	37546	05844	15490	21217
Category	Tops	Tops	Tops	Tops	Tops
Type	Shirts	Shirts	Shirts	Shirts	Shirts
Style	Polo shirts	Polo shirts	Polo shirts	Polo shirts	Polo shirts
Fit	Classic Fit	Tailored Fit	Slim Fit	Relaxed Fit	Classic Fit
Color	Cream	Tan	Blue	Multi	Green
Sleeve	Short Sleeve	Short Sleeve	Short Sleeve	Short Sleeve	Long Sleeve
Graphics	No	No	No	No	No
Size	M	XL	L	L	S
Budget	\$100-\$120	\$70-\$80	\$120-\$130	\$40-\$50	\$80-\$100
Date Needed By	06/31	06/31	07/31	08/15	08/01

**Product
finish info
& size**

Once logged into the ADC, brands can respond to these requests (most conveniently via an API, from IT standpoint) directly to the personal AI agents using the provided personal AI agents' IDs. The personal AI shopping agent can collect the brands' responses and then present the selected options to its human sponsor for final purchase approval.

The quality of data in the ADC will be highly dependent on the real-time updates from the personal AI agents. To provision for that, the incoming requests will initially be logged as "open requests" with a desirable close date after which they will get designated in one of 3 ways: "requests that translated into orders", "unmet demand requests", and "cancelled requests" (Table 3). These designations are very important because for the purposes of demand planning, the "requests that translated into orders" and "unmet demand requests" (for new demand discovery) carry a lot more importance on which to

base actual production signals for the next seasons. The ADC will keep those requests on permanent 24-month basis (to capture all seasons twice over for any seasonal smoothing purposes). This data retention policy would also help manage the size of the ADC, preventing it from becoming an ever-growing repository by putting a time limit to what will be kept in this system of record.

The “unmet demand requests” designation would also serve as an opportunity to address product fairness and justice for the consumer groups that were previously excluded from representation in the demand planning process. If sufficient number of requests are made (representing some critical mass threshold) for Plus Size clothing of a particular type, for example, this will send a strong signal to the brands that there is actual demand, which had been hidden as a blind spot in their rear-view mirror before. Thus, the ADC can enable more product selections to be offered to minority groups with time.

Table 3: Status of requests in the ADC



A working prototype of the ADC could be viewed on the Lovable app at <https://adc-prototype.lovable.app>

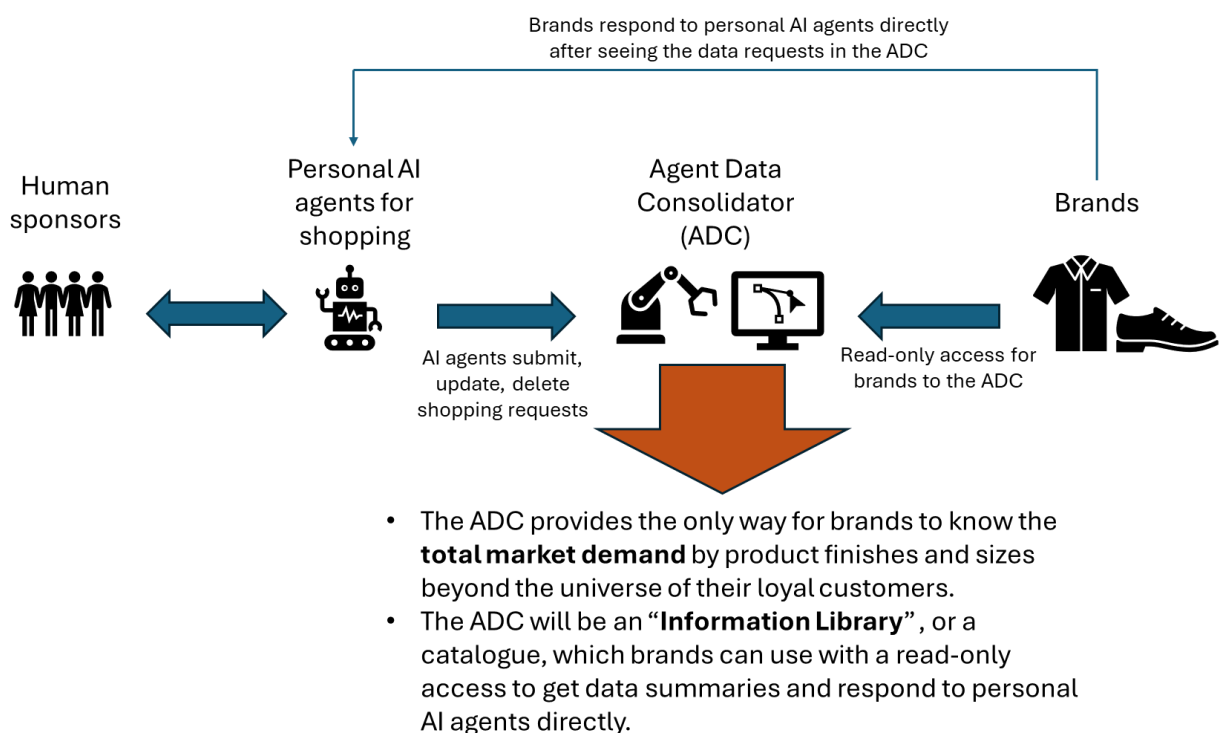
The ADC would in nature be a cloud-based, virtual infrastructure solution within the agentic AI ecosystem. Its biggest dependency will be on the behavior of the personal AI agents, which will be accountable to submit, update, and delete shopping requests from their human sponsors. Those may come in very high volumes because of the frequency at which the human sponsors may update their “wish-lists” and shopping intentions to their personal AI agents. Therefore, this dynamic connection between the personal AI agent and the ADC will also require some process controls in the event when a personal AI agent goes

rogue or experiences a jailbreak, as it may contaminate the ADC with fictitious requests or misleading request changes.

One subtle advantage of the ADC is that it will also serve as an implicit library for “planned requests”, filtering out spontaneous shopping decisions that may still occur at a point-of-sale in physical stores. Since one study found that 20-30% of apparel shopping happens through spontaneous, impulsive decisions¹⁴, those are not a very reliable signal for future demand planning as the same impulse may not repeat again. But requests made through the personal AI agents carry *intention*, which are more useful in nature for future demand planning and would carry more weight.

Viewed as a sociotechnical system, the workflow design (Figure 4) illustrates how the ADC can act as a central “Information Library” (or a catalogue) between the Personal AI agents and the brands.

Figure 4: Workflow design for information exchange in this future scenario



This setup is beneficial for the brands because they would be able to analyze if they are addressing adequately the total *real demand* in the market with their apparel merchandise and if they are over- or underindexed on certain product characteristics and sizes that shoppers are looking for. Thus, the brands can deduce if their product offerings are capturing the fair share of the total market and can react to changing aesthetics earlier

than if they monitor the market themselves via trend-spotting or industry reports (which are often published with a delayed effect).

When the ADC aggregates the data in trailing 12- or 24-month view, it allows brands to get a holistic view of demand at the lowest hierarchy level that stems from *real* demand. This is the basis for building more accurate demand forecasts. Brands can also localize demand (down to a postal (zip) code) and start uncovering location-driven aesthetics that human trend-spotters have missed to register. Moreover, brands can also unload the guesswork from their merchandising teams on the allocation of exact product finishes. All in all, the ADC will be of big service to the brands.

If we try to survey the land for some analogous consumer systems of record that have been developed and sit between individual consumers and businesses, we could examine the case of *Pix*, launched in 2020 by the Central Bank of Brazil¹⁵ because of its (1) size, (2) real-time operations, and (3) central function. *Pix* is system of payment transfers (and an instant payment platform) that facilitates transactions from person-to-person, person-to-business, business-to-business, person-to-government, and business-to-government payments. It processed 6 billion transactions per month in 2024¹⁶, exceeding all credit and debit card transactions combined. Its scope is so comprehensive that it is currently used by 76% of the population and is the most frequently payment platform used by 46%¹⁷. All of this is very impressive considering that it evolved only within a 5-year window. *Pix* assigns its users aliases (similar to how the personal AI agents will have IDs in the ADC), which are portable and the users' information stored in the database is encrypted and protected by law¹⁵.

This example illustrates how a collectively-exhaustive system can seamlessly function even when it comes to the most sensitive exchange for humanity – money. *Pix* can only inspire the future design for the ADC, which would not even handle any transactions, but only serve as a centralized collection point for shopping requests.

VI. Futures timeline and governance for this Theory of Change

If we try to imagine what a futures timeline for the emergence, adoption, and harnessing of the capabilities of the personal AI agents and the creation of the ADC would look like, I would like to offer a 10-year timeline (Figure 5) that provides a reasonable technological horizon before we see the transformational effects signaling improved planetary health reflected in the Sustainability reports by the apparel brands.

Figure 5: Timeline and trends for the algorithmic consumer and ADC developments

2026	2028	2030	2032	2034	2036
- Experimentation and first production deployment of AI agents, which are designed to run on PCs. - Awareness and demand for AI PCs starts growing rapidly. By year end, 54% of all manufactured PCs are AI-capable¹⁴.	10% of people (early adopters) start consistently using AI agents for regular shopping. - AI PCs start handling multi-agent workflows towards “agentic computing”.	20% of people consistently use personal AI agents for shopping. - AI PCs become the industry standard. Every laptop has a personal AI agent. - Use of traditional brand apps for shopping dramatically decreases, algorithmic consumers are in marketing focus.	35% of people consistently use personal AI agents for shopping. - The ADC is put in place by the National Retail Federation (NRF) and starts reflecting meaningful volumes of consumer product choices.	50% of people consistently use personal AI agents for shopping. - Data from the ADC is now used for demand forecasting and production planning. - First effects from the improvement in demand forecasting are reflected in apparel brands’ Sustainability Reports (as well as their Annual Financial Reports)	ADC becomes the industry standard data resource pool for demand planning

Note: In this futures narrative, the outlined developments will hold for OECD countries first

In this Theory of Change, the anticipated outcome would rest on the expectation that in 10 years’ time, at least 50% of people in OECD countries would become algorithmic consumers in order for the ADC to gain meaningful volumes of data that can be used for demand planning at the lowest level in the product hierarchy. I consider this a very plausible scenario because research shows¹⁸ that by the end of 2026, 55% of all personal computers will be AI PCs and by 2030, all PCs will be quipped with personal AI agents¹⁹, which still gives 6 more years till 2036 for humans to adapt, change habits, and start using personal AI agents for shopping on regular basis. If we model this sociotechnical trend to match the pace at which consumers switched to ecommerce shopping²⁰, we see that within any six years, volume of ecommerce has at least doubled with growth in the early years 2014-2020 reaching 400%. If we see similar rates of conversion to apparel shopping via personal AI assistants, the ADC will certainly reach critical mass volume of data to fulfil its purpose.

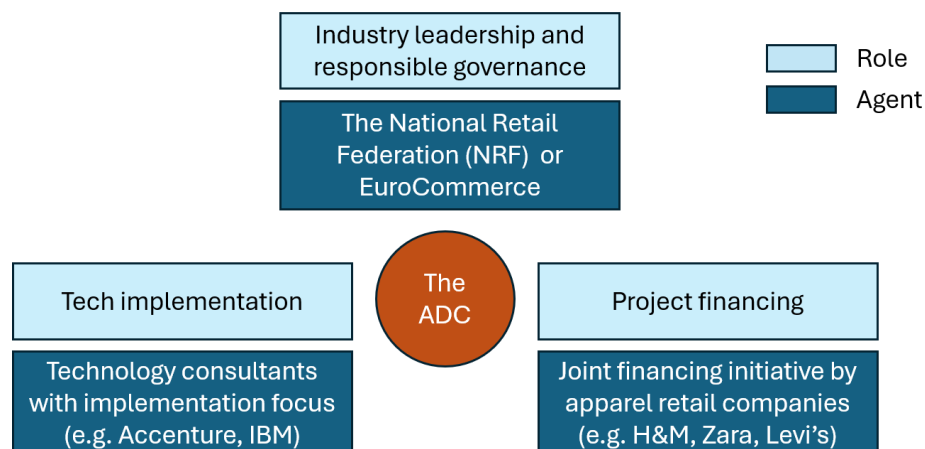
Returning to the analogous case of Pix, one huge factor for its success is rooted in its governance model where the Central Bank of Brazil is the sponsoring body for the implementation and the security guardrails, providing government-mandated support for adoption. With the most eminent government financial institution behind this initiative, Pix rides on the principles of responsible innovation done for public good and not necessarily for private profit. Pix shows that where the governance model is situated is a key element for the success of a massive data centralized system.

In the case of the ADC, the governance model will not be situated within a government body (as no government has direct interest to refine the demand planning practices of private apparel companies), but it could find sponsorship from influential non-government, non-profit organizations. One such organization in the United States could be

the National Retail Federation (NRF), the world’s largest retail trade organization, whose motto is that “Every day, we passionately stand up for the people, policies and ideas that help retail succeed²¹”. The NRF is traditionally focused on career-facilitation and community-building within the retail industry so the ADC could align with the NRF’s purpose to be a connector between people and brands, this time for demand planning. An equivalent organization in the EU would be EuroCommerce, the Brussels-based organization bringing together national retail associations, sector associations, and major retail companies.

The NRF or EuroCommerce can provide the trade connections and thought leadership for the creation of the ADC as an industry solution for demand planning, but they will not be successful if they do not partner with a large tech organization for the technical buildout and management of the ADC. There are multiple tech implementation vendors (such as Accenture or IBM) that could operationalize the ADC and they could be chosen through a comprehensive RFP selection process. But as with any new business venture, the core question will revolve around financing. Finding financing for the creation of the ADC could be a whole business initiative on its own, but here is where the NRF or EuroCommerce could use their influential, industry-gluings status to create a co-funded call for action in which the retail companies in the industry raise and allocate the funds for the ADC creation as they will be the biggest beneficiaries in this Big Data project (Figure 6).

Figure 6: The operational model for the ADC



Thus, the ADC will be a 3-way venture because neither side has competency to create the ADC alone. This will bring complexity and execution risks in this sociotechnical system, which is why the ADC should be understood as a high-effort solution which will necessitate considerable planning, alignment, and resource-allocation between the

agents. The ultimate governance of it will sit with the NRF or EuroCommerce because they will be the least biased agents in the system and can centrally manage this industry solution.

VII. Planetary benefits

Once apparel retail companies start using data from the ADC for demand planning refinement, a reasonable target for improvement in demand forecasts could be 3-10% given that the most analytically advanced companies in the sector (e.g. Levi's) already have internal benchmark thresholds of 92% demand accuracy in aggregate.

Such improvement will first show up in the *inventory turnover* business metric as excess production will be reduced and the dollar value of the average inventory held will decrease. A 10% reduction in average inventory can translate to 11% increase in *inventory turnover*, which, over 365 days, would mean that the company will hold 36.5 fewer days of inventory.

$$Inventory\ Turnover = \frac{Cost\ of\ Goods\ Sold\ (\$USD)}{Cost\ of\ Average\ Inventory\ (\$USD)}$$

Once the company reduces its cost for holding inventory by 36.5 days, this frees up working capital to be deployed elsewhere. In Sustainability reports this is often described as “operational efficiency” or “resource efficiency”.

The decrease in average inventory would also translate to less markdown or season-ending sales and a smaller number of items flowing down to Off-Price resellers (e.g. TJ Maxx). Such “product tail reduction” would curb the impulse of everyday consumers to buy items they don’t need from markdown sales and would decrease transportation and handling costs to ship remaining unsold items to Off-Price resellers. In both cases, this will be an indirect win for the environment.

A decrease in average inventory would also impact another closely related metric, *unsold inventory*, which oftentimes sits as a thorn for apparel retail companies because they need to figure out what to do with it if markdown sales or Off-Price channel flushing does not work out. In many cases, *unsold inventory* is shipped to landfill sites or incarceration depots for destruction, which may take decades to fully degrade, piling waste on the environment.

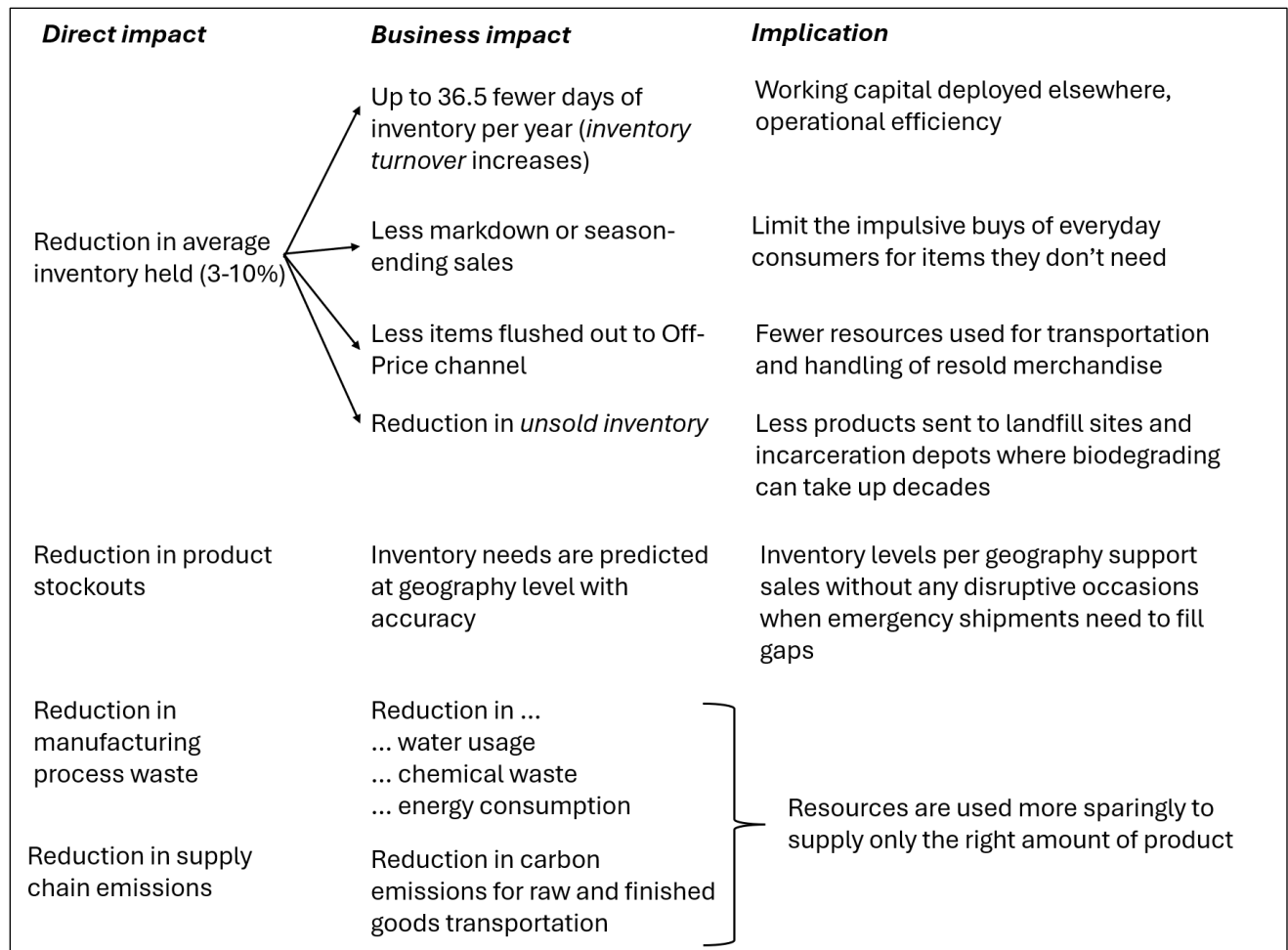
The most egregious case example in the apparel industry on this metric occurred in 2018 at H&M²², which accumulated \$4B worth of unsold inventory, representing 18% of its annual sales. This accumulation was attributed to a long supply chain that overproduced items that customers no longer found desirable or fashionable, which was further aggravated by the fast-paced change of trends in social media, a rising outlet for personal fashion inspiration. To dissolve the inventory, H&M initiated deep markdowns, up to 47%, across stores globally. Consumers were bombarded with items they did not necessarily need, but even this was not enough. To offload excess stock more discreetly, H&M created a discount outlet brand, AFOUND, which was later sold to Secret Sales in 2024. AFOUND allowed H&M to push unsold goods away from their main retail stores.

On the opposite spectrum, understanding total market demand by location (postal code in the ADC) would help reduce *product stockouts* in a given geography. The benefit of this is that emergency shipments would be greatly reduced or eliminated.

Another metric that will be directly positively impacted by improved demand planning and that is frequently found in ESG reports is *carbon emissions* as there will be less such emissions to ship and transfer lower number of containers with raw materials and finished goods. Along with that, we will see a decrease in *water usage*, *chemical waste*, and *energy consumption* in the manufacturing process as companies reduce overproduction.

All of these impacts (Figure 7) reduce the current environmental strain on the planet and lead to a more sustainable future.

Figure 7: Impact of better demand planning on business and ESG metrics



In Sustainability reports of major apparel retail companies, overproduction is not a metric of its own as companies tend to report their sustainability metrics as values derived from multiple initiatives and contributors. However, H&M states that it is decarbonizing its business through “digital tools, better forecasting, smarter inventory²³”, which points to demand planning as a lever that companies can use to achieve that.

Most of the time, as sales grow every year (the goal of any business enterprise), demand forecasts in the most sought-after product categories will be higher every cycle. However, the nuance in this proposal is that demand planning with data from the ADC can help forecasting teams avoid overestimating the quantities. This shield from overestimation is the exact planetary relief that can be realized by running more sparing supply chain operations on global scale.

VIII. Risks and limitations

This Theory of Change depends not only on large-scale sociotechnical trends and developments, but also on specific short- to medium-term business decisions that I find

the need to discuss. The risk of overproduction discussed in this proposal does not solely come from situations where demand planning is overestimating future demand but can also come from other factors that include long production lead times, minimum order quantities, fashion-risk taking, and overall merchandizing strategy.

If the apparel retail company is running on long production lead times, it has to forecast demand oftentimes 6 to over 12 months in advance²⁴. This is the case with most traditional apparel companies (currently holding 82% sales share in the global market²⁵) because their supply chain is not built for quick and cheap, reactive, capsule-style creations in the way fast fashion is. If we look at Levi's for reference, the supply chain there is on the longer side at 14 months (since denim production requires more processing time than other fabrics) and the reality is that there are multiple external forces that can throw off the process that on paper has been designed to run smoothly²⁶. Such external circumstances have in the past included port congestions, labor shortages at operational sites, and shortages in warehousing capacities in various distribution centers. So in practice, lead times in the apparel retail industry are and would remain long (longer than 6 months), which then prompts demand planning teams to add a layer of "safety volume" to the forecasts because of optimistic forward-looking corporate plans and expectations.

This is where the ADC can help alleviate the "unknown factor" in future demand by offering some signals for future demand (grounded in consumer intentions) if the personal AI agents are able to shop around future calendar events in their human sponsors' schedules. So even if long lead times often predispose overproduction, the ADC can still help mitigate that risk to slightly lower overproduction.

Another operational constraint leading to overproduction is the minimum order quantities in which apparel retailers have no choice but to sometimes order above their forecasted quantities. This could be especially detrimental for the smaller consumer groups that this paper has discussed are excluded from the product portfolios, such as the Plus Size consumer. Sometimes the choice there is to either overproduce something or not produce it at all. This is where the ADC can be helpful again because if there is a signal for unmet demand, then more apparel companies would choose to produce rather than not, believing that actual demand will exceed the minimum order quantity.

Another risk for overproduction is when apparel companies play fashion-risk taking in which they design and produce a fashion line that is not typical for their portfolio. The extenuating factor here is that oftentimes such experimental lines involve small quantities or capsule collections, which, in aggregate, do not command a massive production operation. The ADC could also be helpful in such cases of fashion-risk taking by guiding the fashion designers to peg the experimentation process around product attributes (fabric,

silhouette, finish, buttons, patterns, etc) that consumers are requesting in greater number than others.

In parallel to the above, another merchandising strategy that propels apparel retail companies towards overproduction is when the strategy is to emphasize variety for the consumer. In such cases, there will be a long tail of products that will get produced but never bought. Here, the ADC can again be helpful as input in the strategy formulation to rationalize (or cap) the number of product items planned.

In sum, overproduction in the apparel retail industry can be conceptualized as a function of analytical demand forecasting, lead times, minimum order quantities, and merchandising strategy.

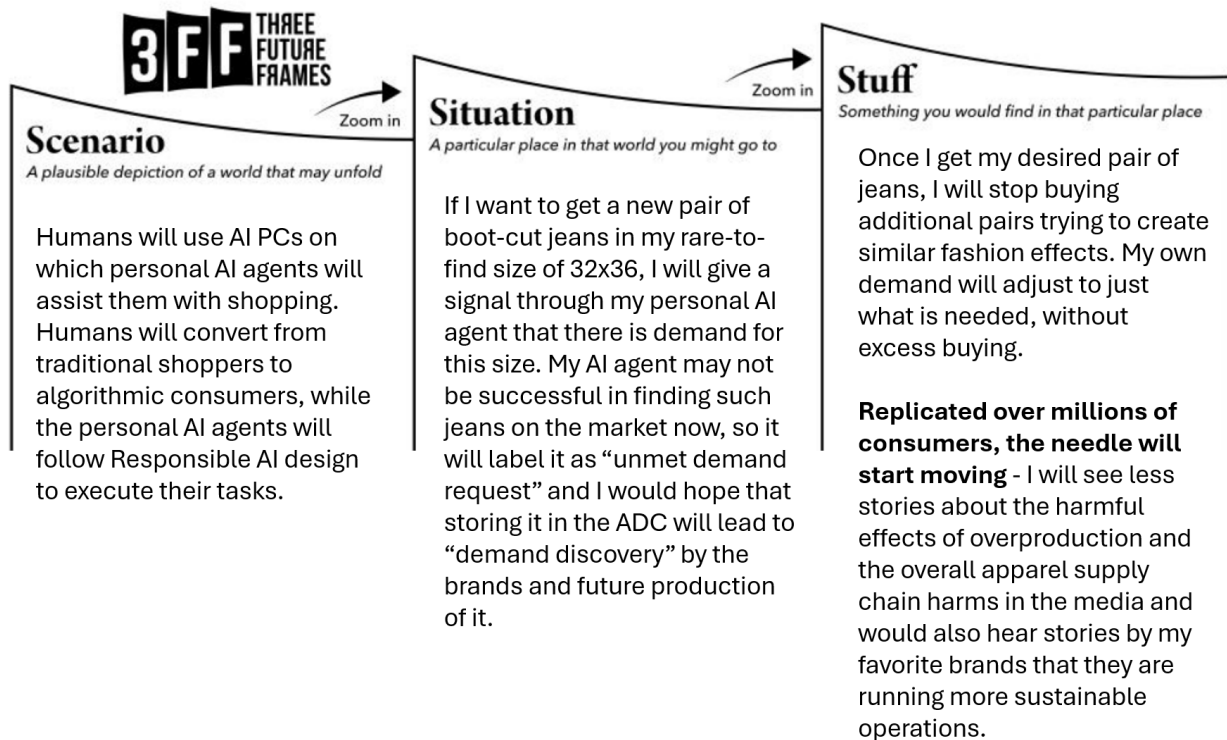
$$\text{Overproduction} = f(\text{demand forecasting, lead times, min order quantity, merch strategy})$$

Data from the ADC introduced in this proposal can directly help companies sharpen their analytical demand forecasts and the effect will be enhanced when the ADC data is used and weighed in the context of the other supply chain constraints and the overall merchandising strategies. This will produce a demand plan that combines both real marketplace demand and the company's own constraints and strategies. The net result will be a decrease in overproduction, sparing unnecessary strain for the environment.

IX. 3 Future Frames: Scenarios, Situations, Stuff

The desired long-term outcome that this paper rests on is planetary health. Now that we have outlined 1) the current practice of demand planning for apparel goods with the specific area where it can be improved, 2) the emergence of the futures new persona (the algorithmic consumer), 3) the creation of the futures new tool (the Agent Data Consolidator), 4) the futures timeline for this Theory of Change, and 5) the positive planetary effects that it can bring, I will use the “3 Future Frames”²⁷ to situate this entire theory into practical illustration from the point of view of the individual consumer (myself in this case).

Figure 8: The 3 Future Frames – personalized example



If we can scale this personal example to comparable future frames for millions of other consumers, I am a firm believer that we could achieve the goals laid out in this futures narrative. What it takes is equal portions:

- (1) Technology solutionism (through the ADC) with a 3-way governance model,
- (2) Responsible AI design (for the personal AI agents), and a
- (3) Socioeconomic transformation

in which the individual consumers switch to agentic commerce and the business analytics community alters the current methodology for demand forecasting utilizing the granularity of data available. The latter third point may be the toughest challenge because it rests on broader socioeconomic trends in which established habits and norms have evolve to new paradigms.

X. Main Audience for this proposal

This proposal would be of utmost interest to the following groups:

- Demand planning leaders at apparel retail companies
- Supply chain and operations leaders at apparel retail companies

- Demand planning teams and supporting personnel
- National Retail Federation (NRF) community program developers
- Agentic AI developers (to consider the Responsible AI design requirements)
- Technology solutionists who recognize the merits of the ADC
- Technology consultancies who can use this strategic blueprint for the creation of the ADC

The key beneficiaries of this proposal would not only be the apparel retail companies (which can improve their P&L through leaner supply chains) but humanity as a whole as the planetary resources will be used more sparingly for industrial production and consumption.

XI. Summary

This proposal developed a futures narrative in which agentic AI can be used to help predictive AI create more accurate demand planning forecasts for apparel goods. With current demand forecasts achieving best-in-class accuracy of 92%, the push will be to sharpen demand planning by additional 3-10% to further cut overproduction and excessive supply. This will produce a measurable effect on planetary health by sparing overproduction and creating a leaner supply chain.

The timeline in this futures narrative spans the next 10 years (2026-2036) and the proposed results will be achieved through a futures chain of events that will see the rise of a new persona (the algorithmic consumer), a new tool (the Agent Data Consolidator), and a new sociotechnical paradigm in which 50% of consumers in OECD countries would switch to using personal AI shopping agents to fulfill their *intentional* shopping needs.

For the rise of the new personal, AI developers will need to ensure that a certain set of Responsible AI principles is embedded in each personal AI agent. For the new tool creation, the Agentic Data Consolidator, an industry-wide project has to be put in place to raise financing, govern, and realize this vision. The ADC would be built on the principles of technology solutionism capturing human behavior (apparel shopping needs) and aggregating that data into production signals for apparel companies to use in demand planning. The differentiating aspect of the ADC is the fact that it represents total market demand and can highlight blind spots that brands have missed noticing and planning for, which in turn will give visibility to underserved groups, such as the Plus Size consumers. The ADC will avoid becoming a surveillance tool by taking only attribute information and no personal identifiable data. Analogous examples, such as Brazil's Pix payment system, already exist to serve as guidance and inspiration.

Once brands can incorporate the data from the ADC into their demand planning, they can create more accurate plans at the lowest hierarchy product level and address needs by geography. This will translate into reductions in environmentally harmful practices in the supply chains and will relieve the planet from unnecessary production strain. Thus, this proposal offers a chance for humanity to leverage the advent of agentic AI for the benefit of the planet.

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